LOOP 2: ONE ALGEBRA MOVE PER LINE

1. Write signed KVL $6 - 5I_2 + 2I_3 = 0$
2. Substitute I_3 $6 - 5I_2 + 2(I_1 - I_2) = 0$
3. Distribute $+2$ $6 - 5I_2 + 2I_1 - 2I_2 = 0$
4. Reorder terms $6 + 2I_1 - 5I_2 - 2I_2 = 0$
5. Combine like terms $6 + 2I_1 - 7I_2 = 0$
6. Subtract 6 $2I_1 - 7I_2 = -6$
7. Multiply by -1 $\boxed{-2I_1 + 7I_2 = 6}$